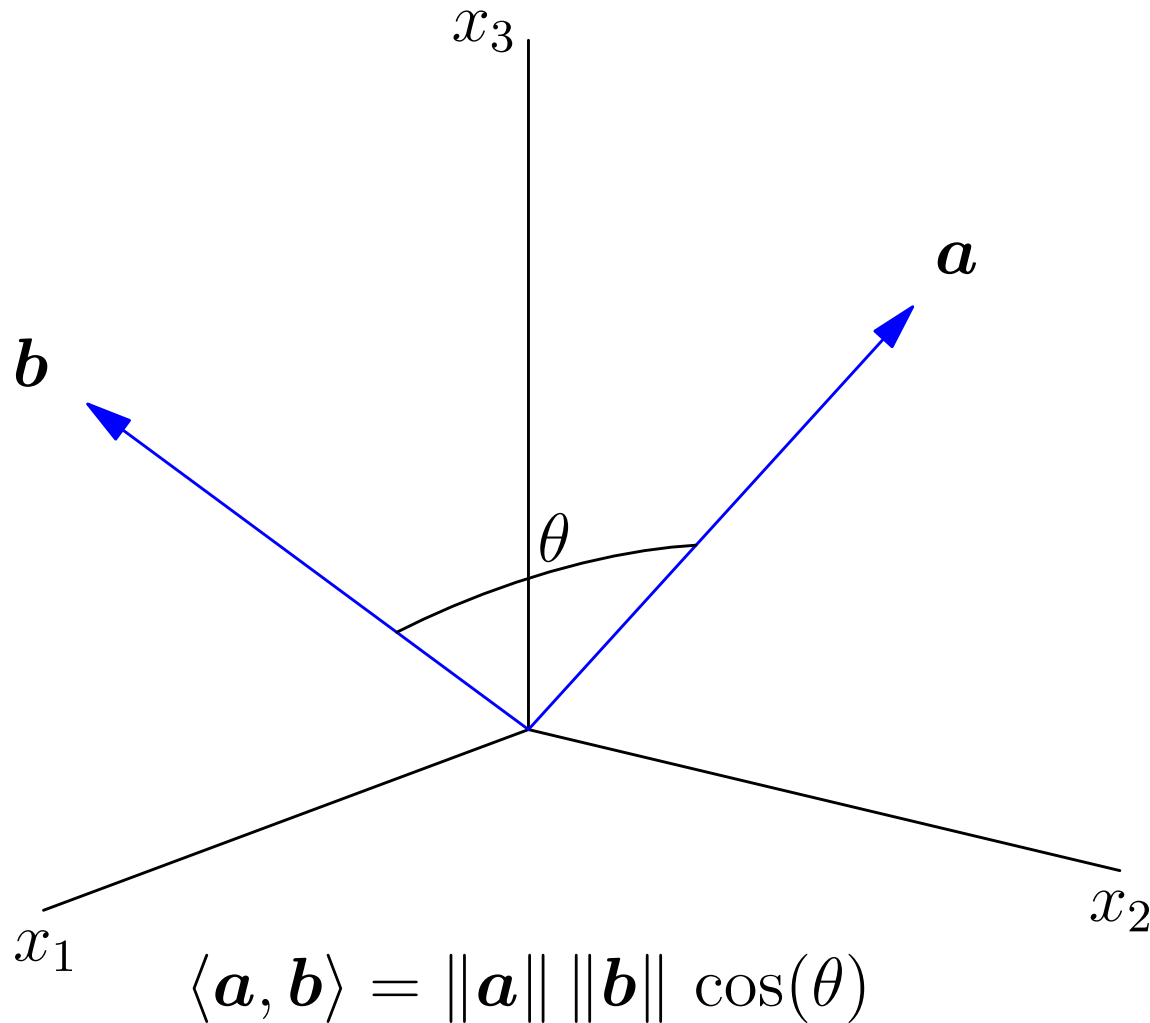


Mathematics of Machine Learning

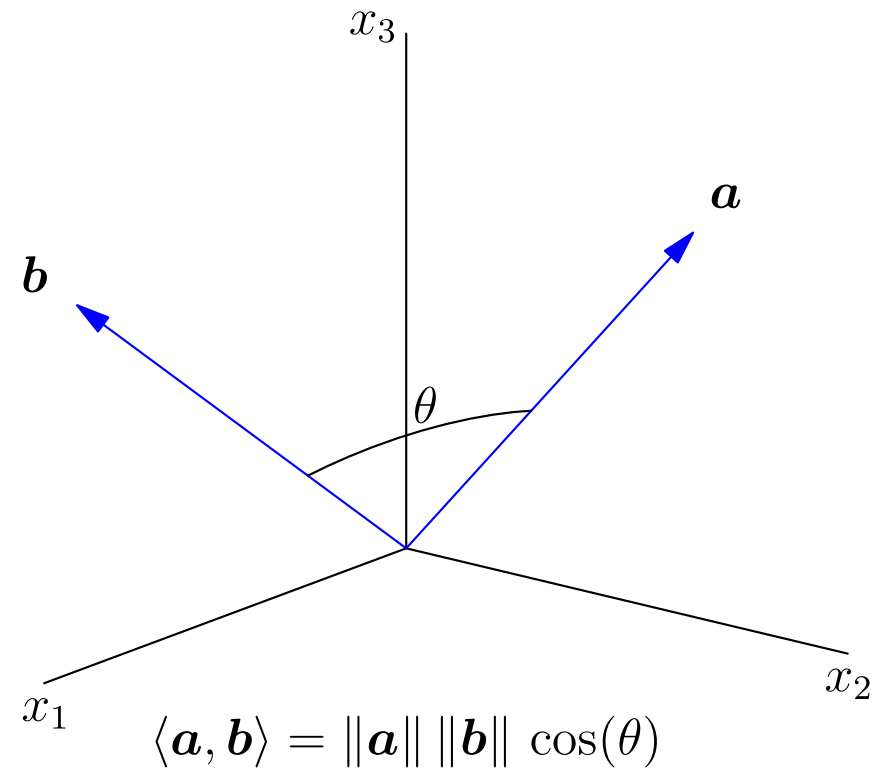
Inner Product Spaces



Inner products, operators

Outline

1. Inner Products
2. Operators



Recap

- We have looked at vector spaces
- Recall that vector spaces don't just apply to normal vectors ($\mathbb{R}^n$), but to matrices, functions, sequences, random variables, . . .
- Proper distances or metrics, $d(\mathbf{x}, \mathbf{y})$, allow us to construct ideas about geometry of the vector space
- Norms, $\|\mathbf{x}\|$, allow us to reason about the size of a vector
- Norms induce a distance, $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$

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Inner Products

- We will often consider objects with an *inner product*, $\langle, \rangle$
- For vectors in $\mathbb{R}^n$

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i$$

- For functions

$$\langle f, g \rangle = \int_{x \in \mathcal{I}} f(x) g(x) dx$$

- For $m \times n$ matrices

$$\langle \mathbf{A}, \mathbf{B} \rangle = \text{Tr} \mathbf{A}^\top \mathbf{B} = \sum_{i=1}^m \sum_{j=1}^n A_{ij} B_{ij}$$

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Axioms of Inner Products

- An inner product satisfies

1. $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$ for all $\mathbf{x} \in \mathcal{V}$
2. $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ if and only if $\mathbf{x} = \mathbf{0}$
3. $\langle \alpha \mathbf{x}, \mathbf{y} \rangle = \alpha \langle \mathbf{x}, \mathbf{y} \rangle$
4. $\langle \mathbf{x}, \mathbf{y} + \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{z} \rangle$
5. $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$

- We can show that $\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$ satisfies the axioms of a norm, so that an inner-product space is a normed space
- The norm associated with the inner-product for vectors in $\mathbb{R}^n$ (i.e. $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y}$) is the Euclidean norm $\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}$

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Cauchy-Schwarz Inequality

- One of the most important results of inner-product spaces, known as the **Cauchy-Schwarz inequality** is that

$$\langle \mathbf{x}, \mathbf{y} \rangle^2 \leq \langle \mathbf{x}, \mathbf{x} \rangle \langle \mathbf{y}, \mathbf{y} \rangle = \|\mathbf{x}\|^2 \|\mathbf{y}\|^2$$

- Or

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|$$

- This is a very general result so for example

$$\left| \int f(x)g(x)dx \right| \leq \sqrt{\left(\int f^2(x)dx \right) \left(\int g^2(x)dx \right)}$$

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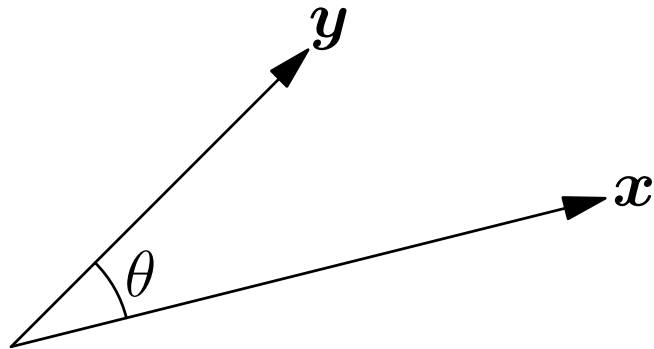
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Angles Between Vectors

- A natural interpretation of the inner product is in providing a measure of the angle between vectors



$$\langle x, y \rangle = x^T y = \|x\| \|y\| \cos(\theta)$$

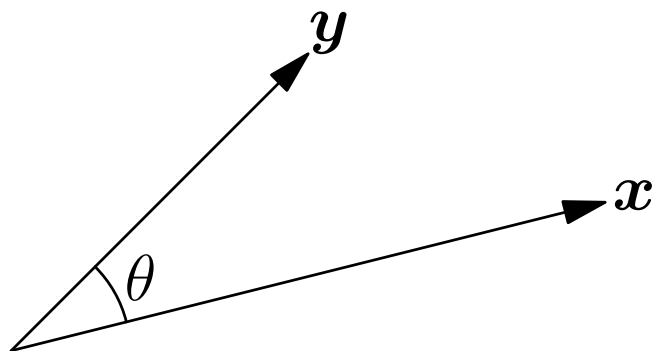
- Vectors are orthogonal if $\langle x, y \rangle = 0$
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- Note that $\sin(x)$ and $\cos(x)$ are orthogonal in the interval $[0, 2\pi]$

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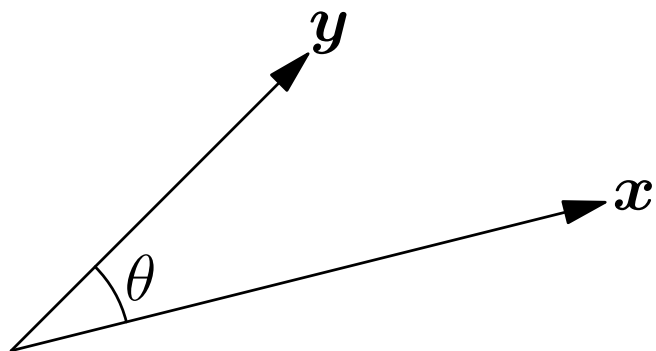
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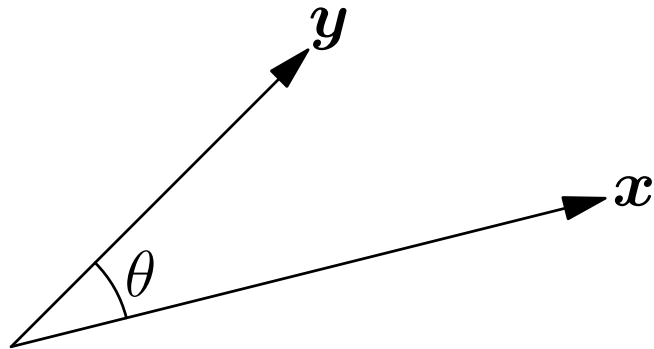
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Basis Functions

- Any set of vectors $\{\mathbf{b}_i | i = 1, \dots\}$ that span the space can be used as a basis or coordinate system
- The simplest and most useful case is when the vectors are orthogonal and normalised (i.e. $\|\mathbf{b}_i\| = 1$)

- In $\mathbb{R}^3$ we could use $\mathbf{b}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\mathbf{b}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $\mathbf{b}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

- This is not unique as we can rotate our basis vectors
- For an orthonormal basis we can write any vector as

$$\hat{\mathbf{x}} = \begin{pmatrix} \mathbf{x}^\top \mathbf{b}_1 \\ \mathbf{x}^\top \mathbf{b}_2 \\ \mathbf{x}^\top \mathbf{b}_3 \end{pmatrix}$$

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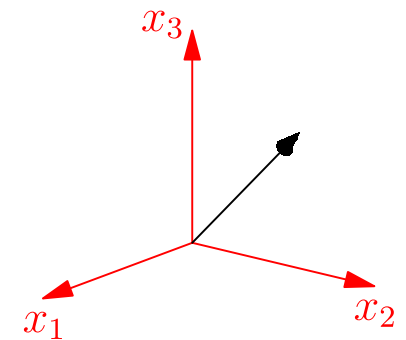
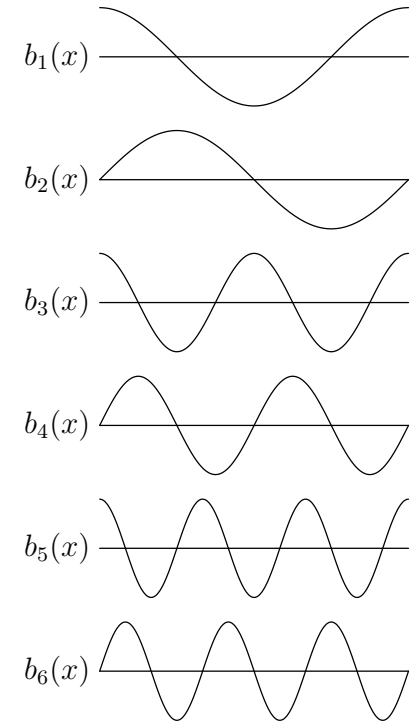
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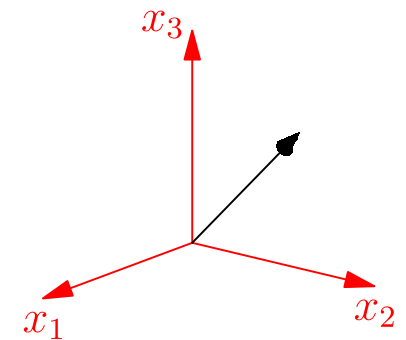
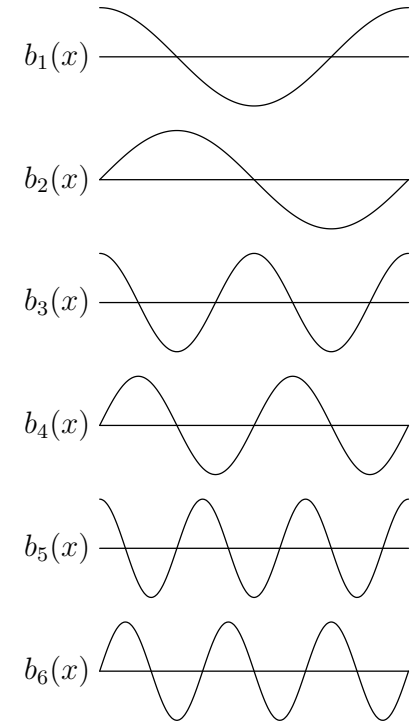
Orthogonal Functions

- For functions we can use any ortho-normal set of functions as a basis
- The most familiar are the Fourier functions $\sin(n\theta)$ and $\cos(n\theta)$
- Any function in $C(0,2\pi)$ can be represented by a point $\mathbf{f} = \begin{pmatrix} \langle f(x), b_0(x) \rangle \\ \langle f(x), b_1(x) \rangle \\ \vdots \end{pmatrix}$
- There might be an infinite number of components
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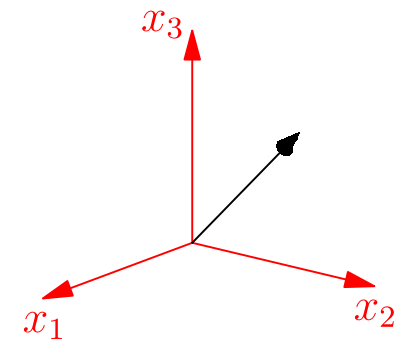
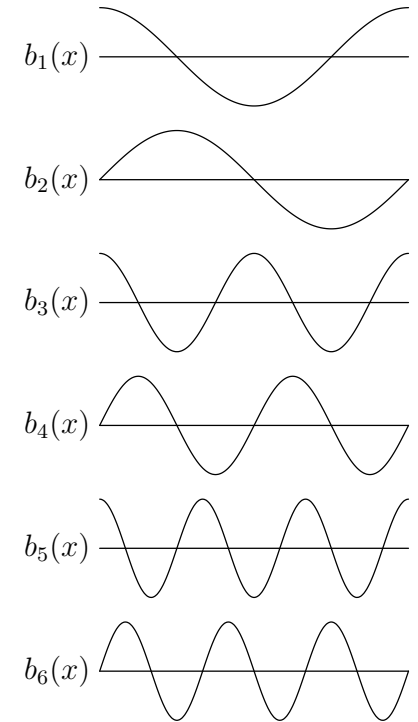
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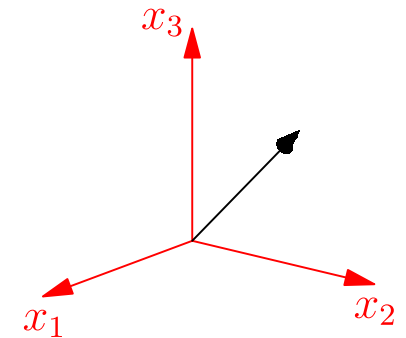
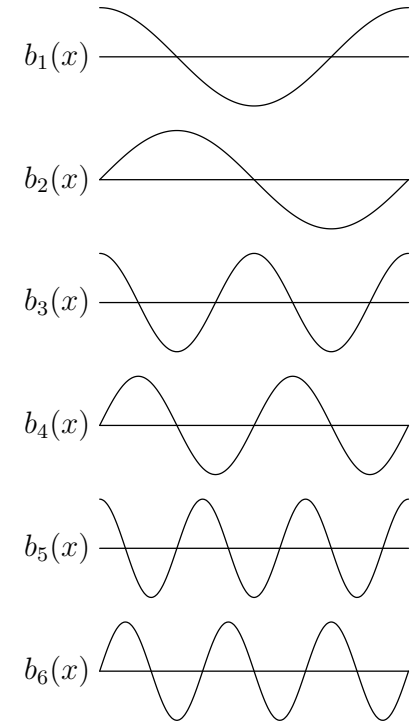
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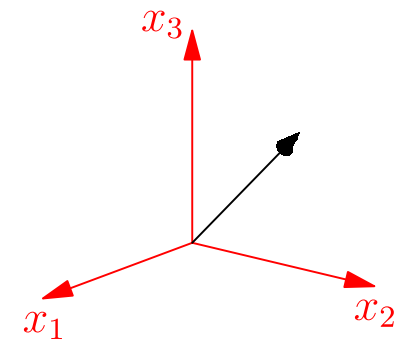
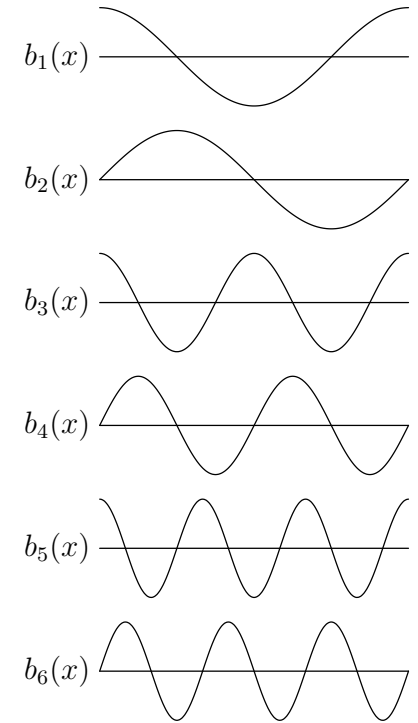
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- Any function in $C(0,2\pi)$ can be represented by a point $\mathbf{f} = \begin{pmatrix} \langle f(x), b_0(x) \rangle \\ \langle f(x), b_1(x) \rangle \\ \vdots \end{pmatrix}$
- There might be an infinite number of components
- This is analogous to points in $\mathbb{R}^n$ (for large n)



Projection Operator

- Given a vector a we can project it onto a vector b

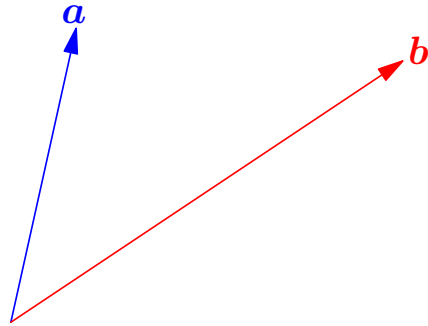
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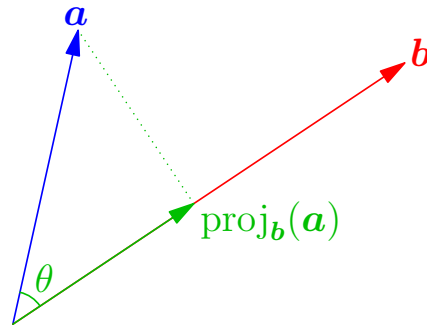
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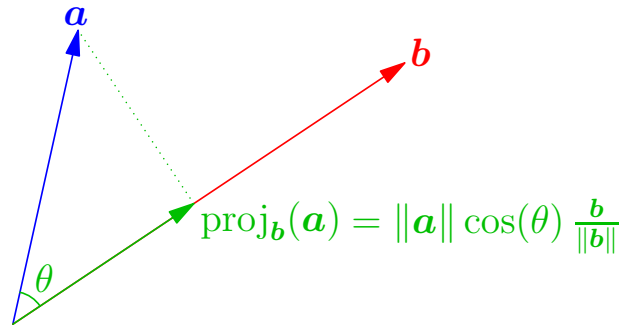
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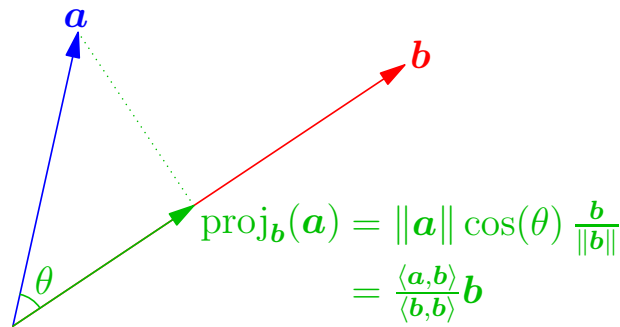
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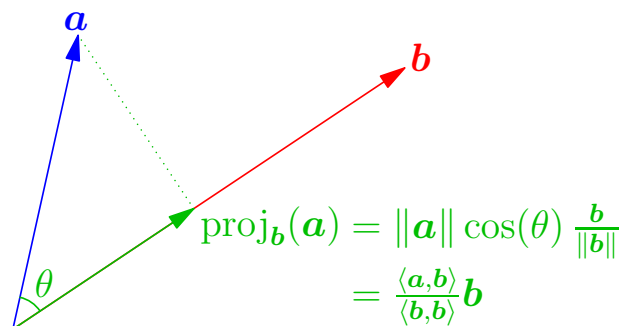
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- Using $\langle a, b \rangle = \|a\| \|b\| \cos(\theta)$ and $\langle b, b \rangle = \|b\|^2$

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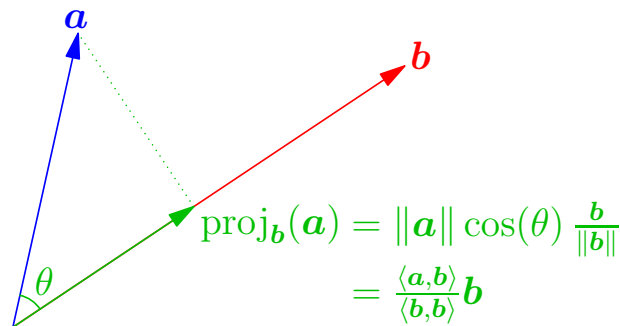


- Using $\langle \mathbf{a}, \mathbf{b} \rangle = \|\mathbf{a}\| \|\mathbf{b}\| \cos(\theta)$ and $\langle \mathbf{b}, \mathbf{b} \rangle = \|\mathbf{b}\|^2$
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$$\text{proj}_{\mathbf{b}}(\text{proj}_{\mathbf{b}}(\mathbf{a})) = \text{proj}_{\mathbf{b}}\left(\frac{\langle \mathbf{a}, \mathbf{b} \rangle}{\langle \mathbf{b}, \mathbf{b} \rangle} \mathbf{b}\right)$$

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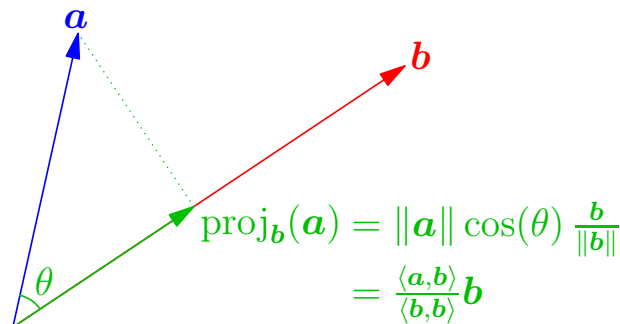


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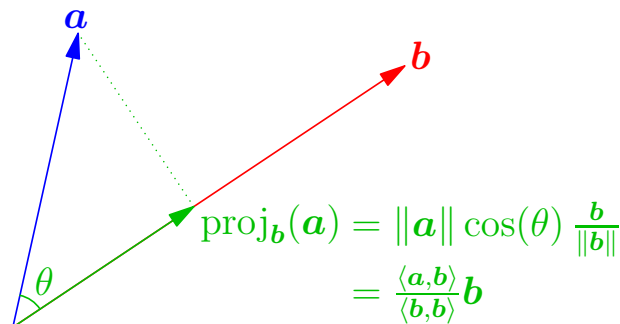


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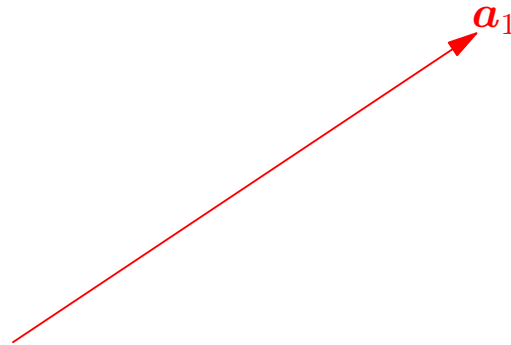
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Constructing an Orthonormal Basis

- Given a set of linearly independent vectors $(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n)$ we can construct an orthonormal basis

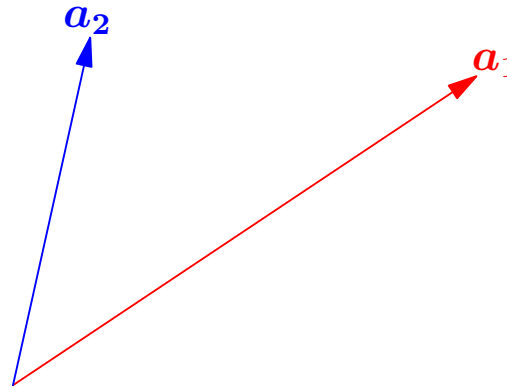
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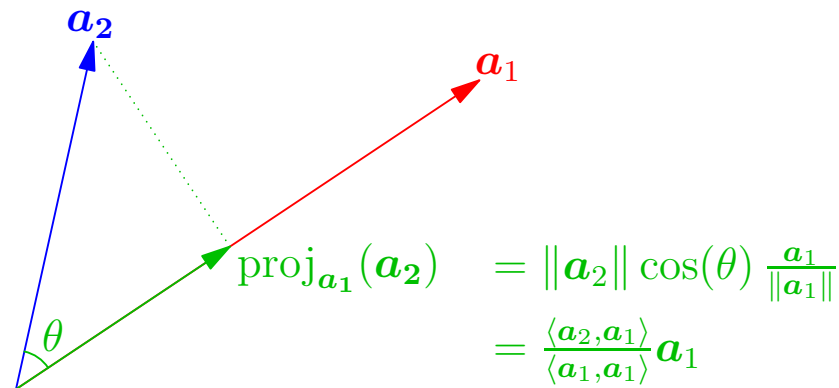
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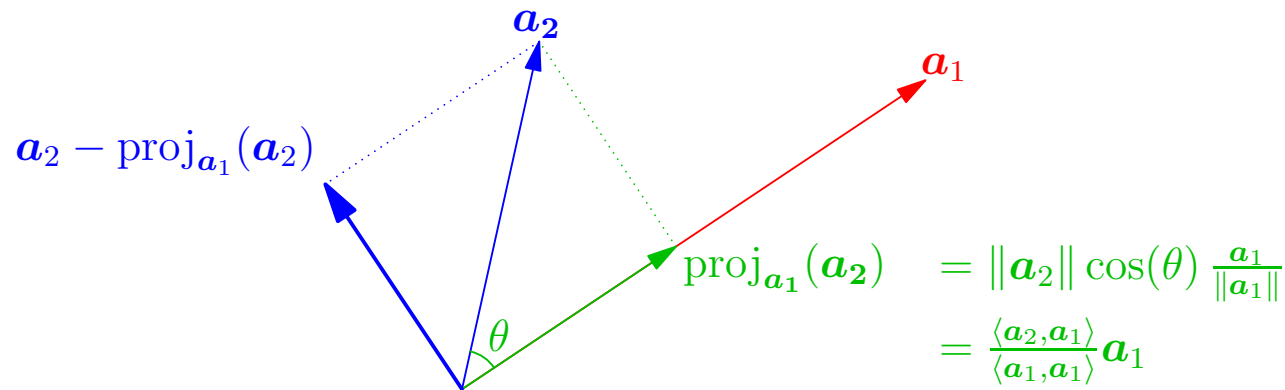
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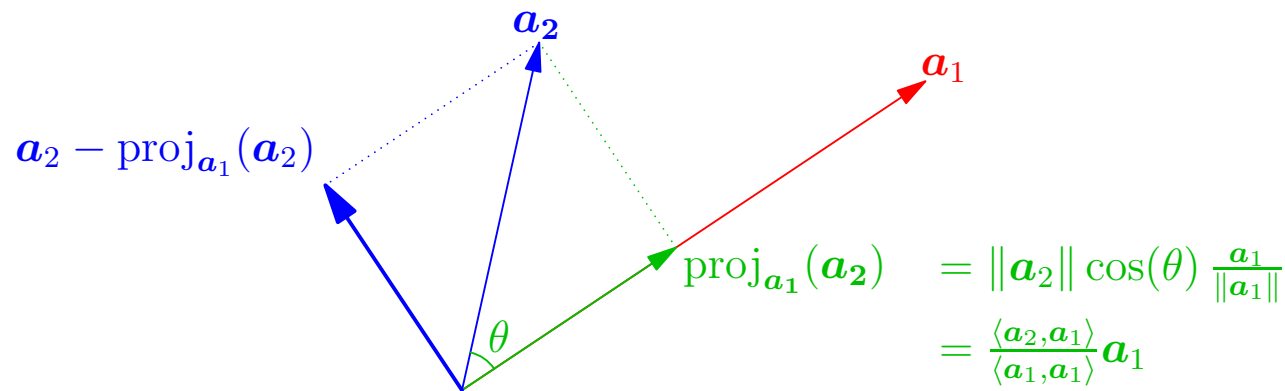
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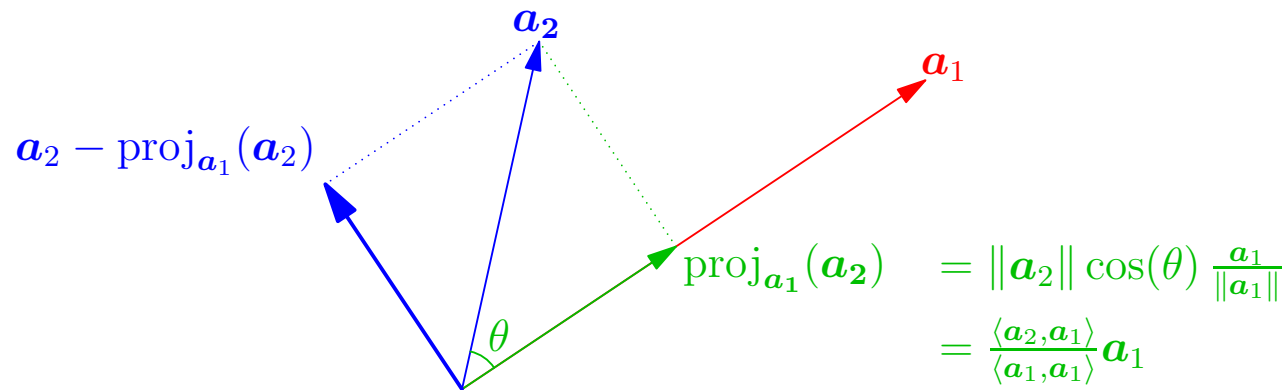
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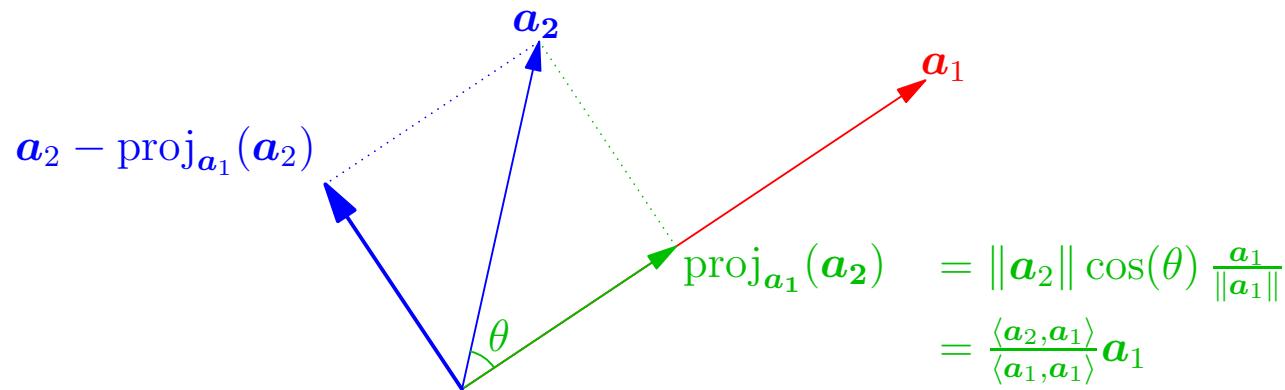
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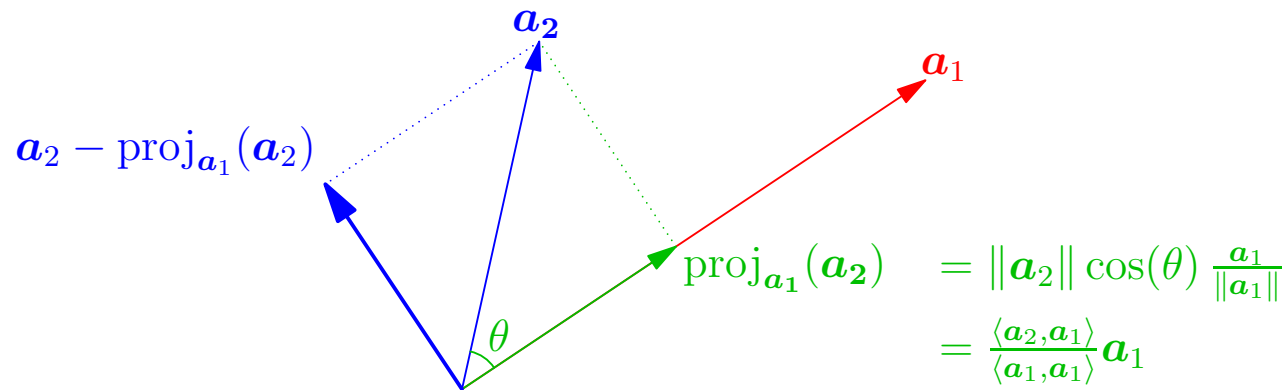
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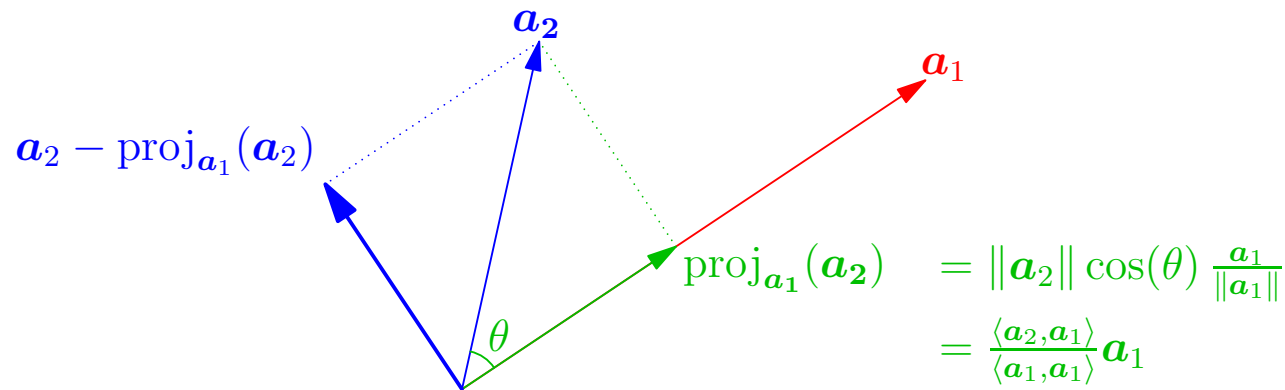
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- The algorithm in the last slide is known as **Gram-Schmidt**
- We can perform it on any inner-product space
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- We have gone to these lengths as we want to show that many properties of vectors are shared by other objects (matrices, functions, etc.)
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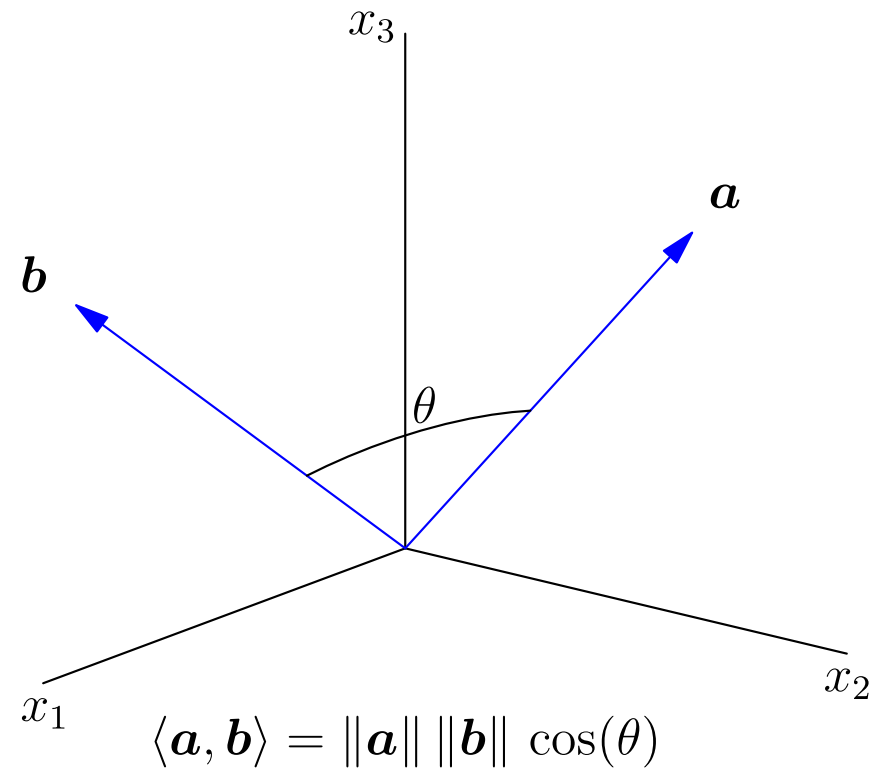
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Outline

1. Inner Products
2. **Operators**



Operators

- In machine learning we are interested in transforming our input vectors into some output predictions
- To accomplish this we will apply some mapping or operators on the vector $\mathcal{T} : \mathcal{V} \rightarrow \mathcal{V}'$
- This says that $\mathcal{T}$ maps some object $x \in \mathcal{V}$ to an object $y = \mathcal{T}[x]$ in a new vector space $\mathcal{V}'$
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Linear Operators

- Operators are in general very complicated, but a particularly nice set of operators are linear operators
- $\mathcal{T}$ is a linear operator if
 1. $\mathcal{T}[a\mathbf{x}] = a\mathcal{T}[\mathbf{x}]$
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Matrix multiplication

- For an $\ell \times m$ matrix $\mathbf{A}$ and an $m \times n$ matrix $\mathbf{B}$ we can compute the $(\ell \times n)$ product, $\mathbf{C} = \mathbf{AB}$, such that

$$C_{ij} = \sum_{k=1}^m A_{ik} B_{kj} \quad \left(\begin{array}{|c|} \hline \text{---} \\ \hline \text{---} \\ \hline \text{---} \\ \hline \end{array} \right) \left(\begin{array}{|c|c|c|} \hline | & | & | \\ \hline | & | & | \\ \hline | & | & | \\ \hline \end{array} \right) = \left(\begin{array}{|c|c|c|} \hline \blacksquare & \blacksquare & \blacksquare \\ \hline \blacksquare & \blacksquare & \blacksquare \\ \hline \blacksquare & \blacksquare & \blacksquare \\ \hline \end{array} \right)$$

- Treating the vector $\mathbf{x}$ as an $n \times 1$ matrix then

$$\mathbf{y} = \mathbf{Ax} \quad \Rightarrow \quad y_i = \sum_j A_{ij} x_j \quad \left(\begin{array}{|c|} \hline \text{---} \\ \hline \text{---} \\ \hline \text{---} \\ \hline \end{array} \right) \left(\begin{array}{|c|} \hline | \\ \hline | \\ \hline | \\ \hline \end{array} \right) = \left(\begin{array}{|c|} \hline \blacksquare \\ \hline \blacksquare \\ \hline \blacksquare \\ \hline \end{array} \right)$$

- Using the same matrix notation we can define the inner product as

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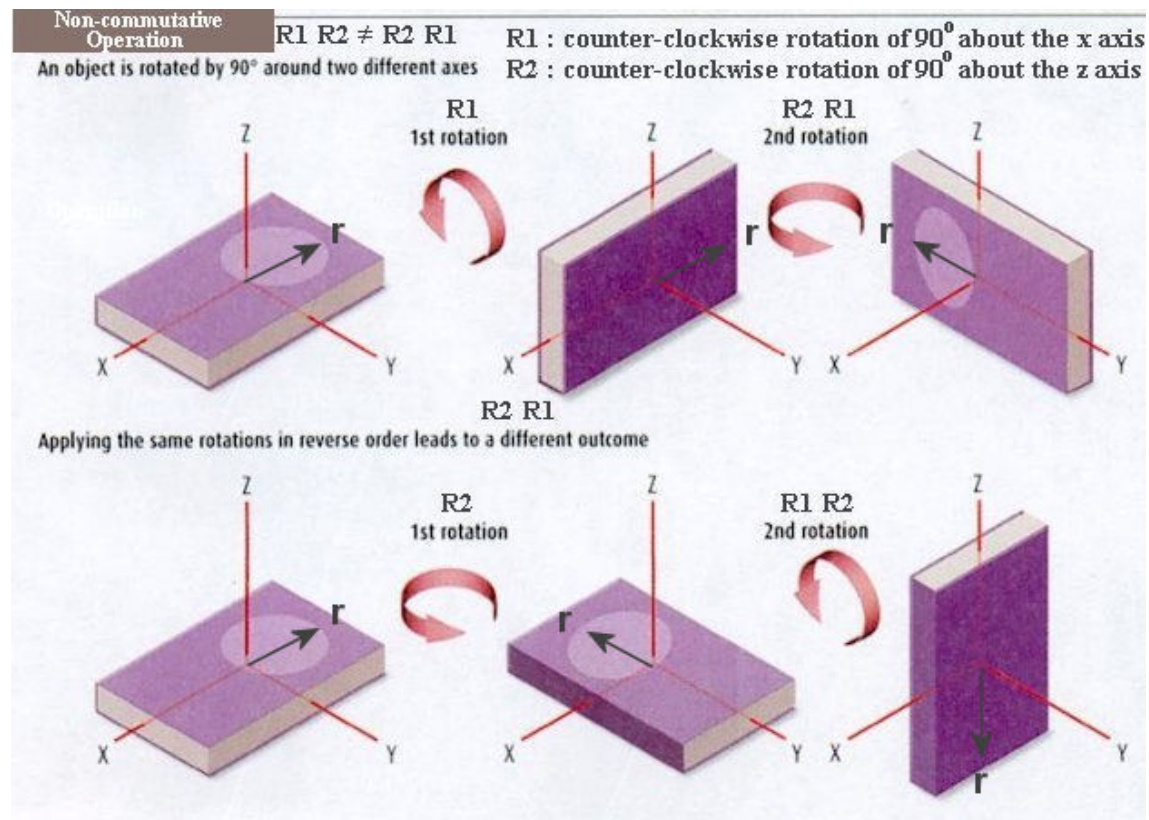
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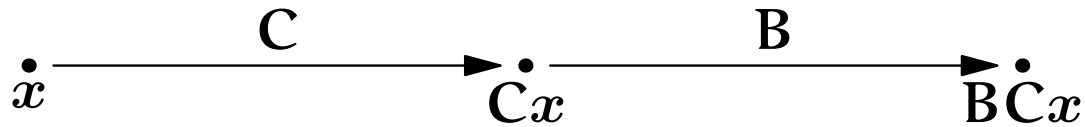
Associativity of Mappings

$\dot{x}$

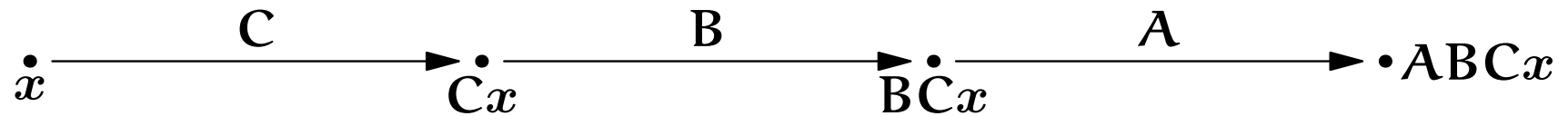
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$$\dot{x} \xrightarrow{C} \dot{C}x$$

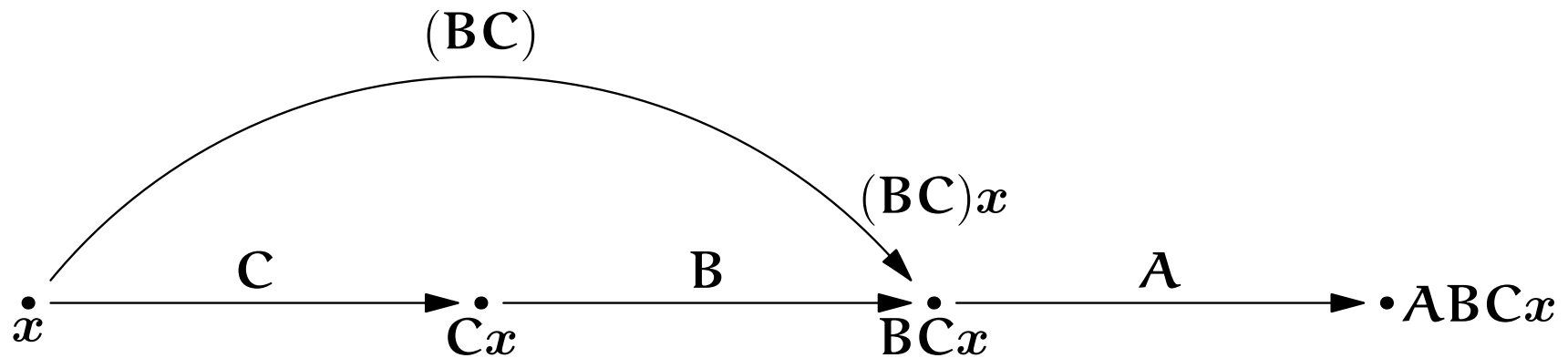
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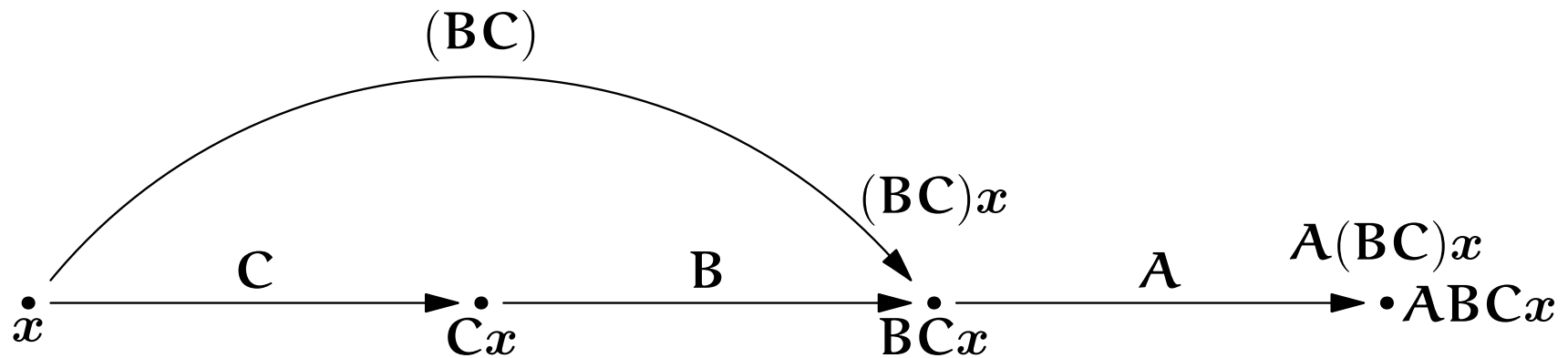
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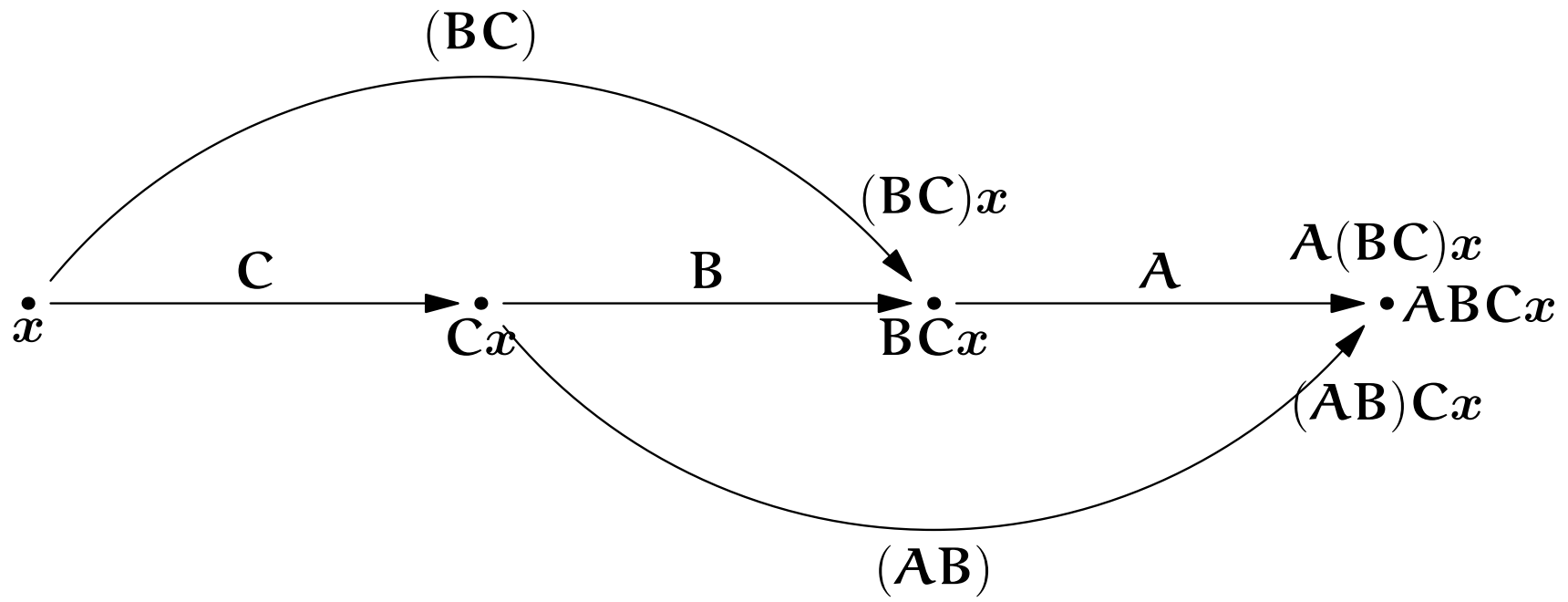
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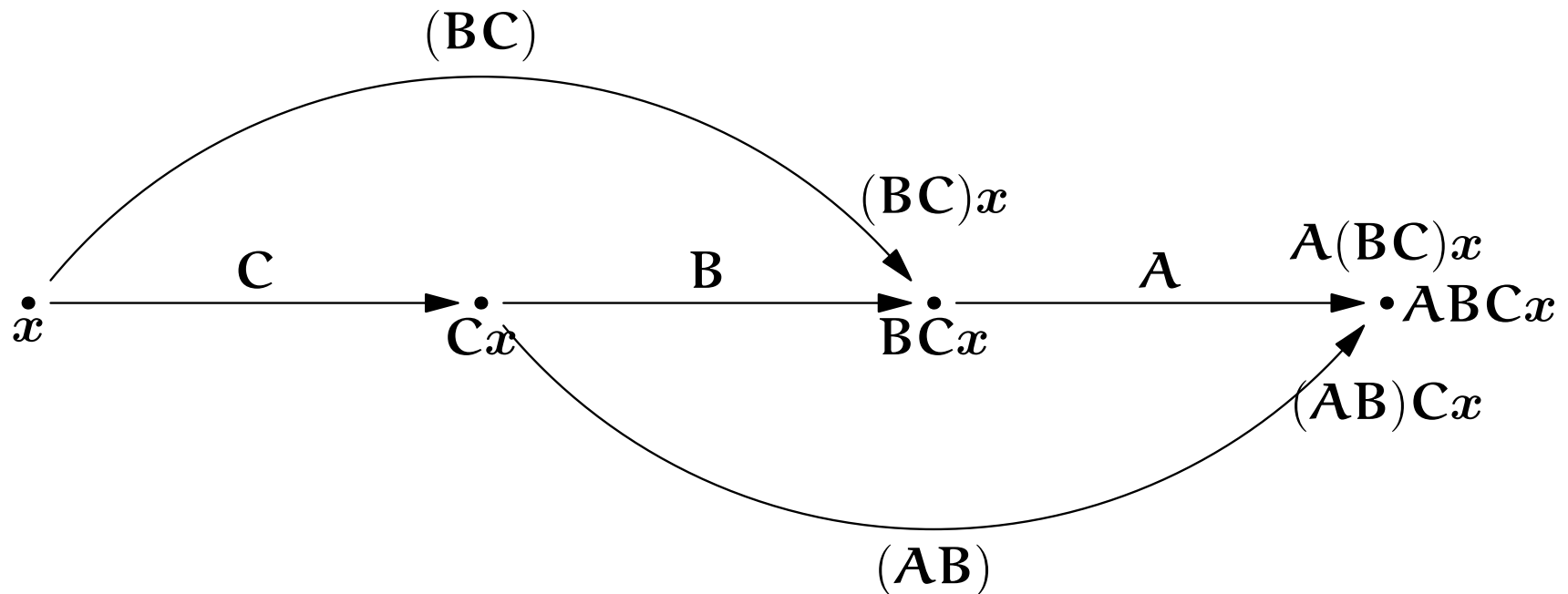
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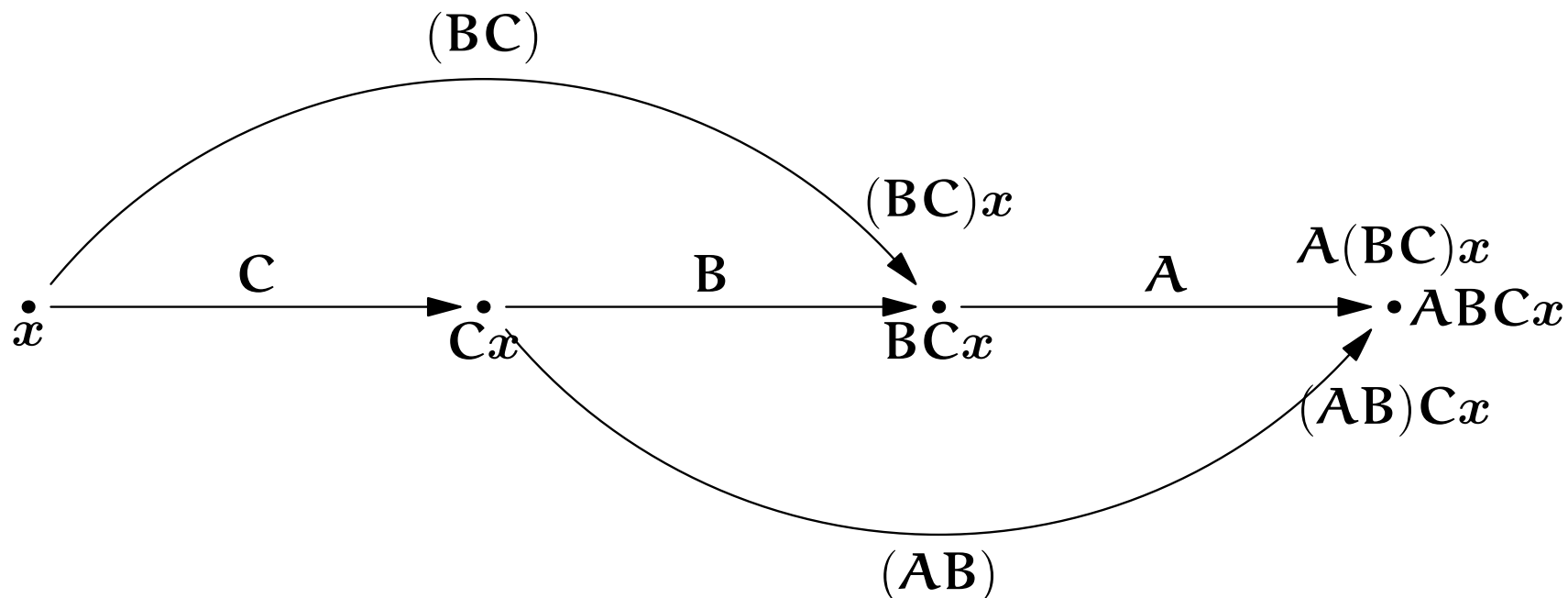


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- This implies $A(BC) = (AB)C$

Kernels

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$$g(x) = \mathcal{T}[f] = \int_{y \in \mathcal{I}} K(x, y) f(y) dy$$

- Our domain does not need to be one dimensional, e.g.

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Kernels in Machine Learning

- Kernels are used heavily in machine learning
- In kernel methods such as SVM, SVR, Kernel-PCA
- They are also used in Gaussian Processes
- In all these cases we consider symmetric, positive semi-definite kernels
- Sometimes they can be interpreted as covariance between random functions

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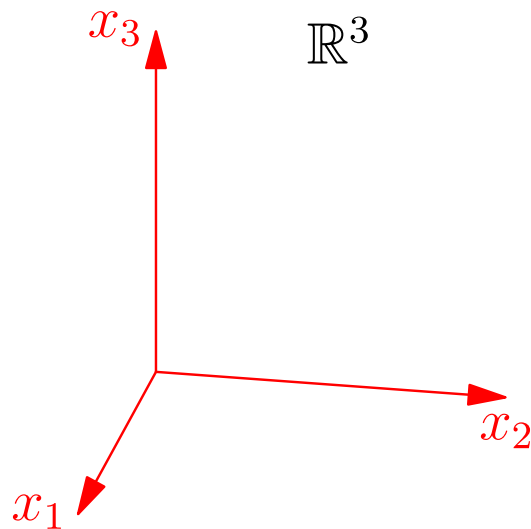
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- In general a linear operator will map vectors between different vector spaces
- E.g. $\mathbb{R}^3 \rightarrow \mathbb{R}^2$

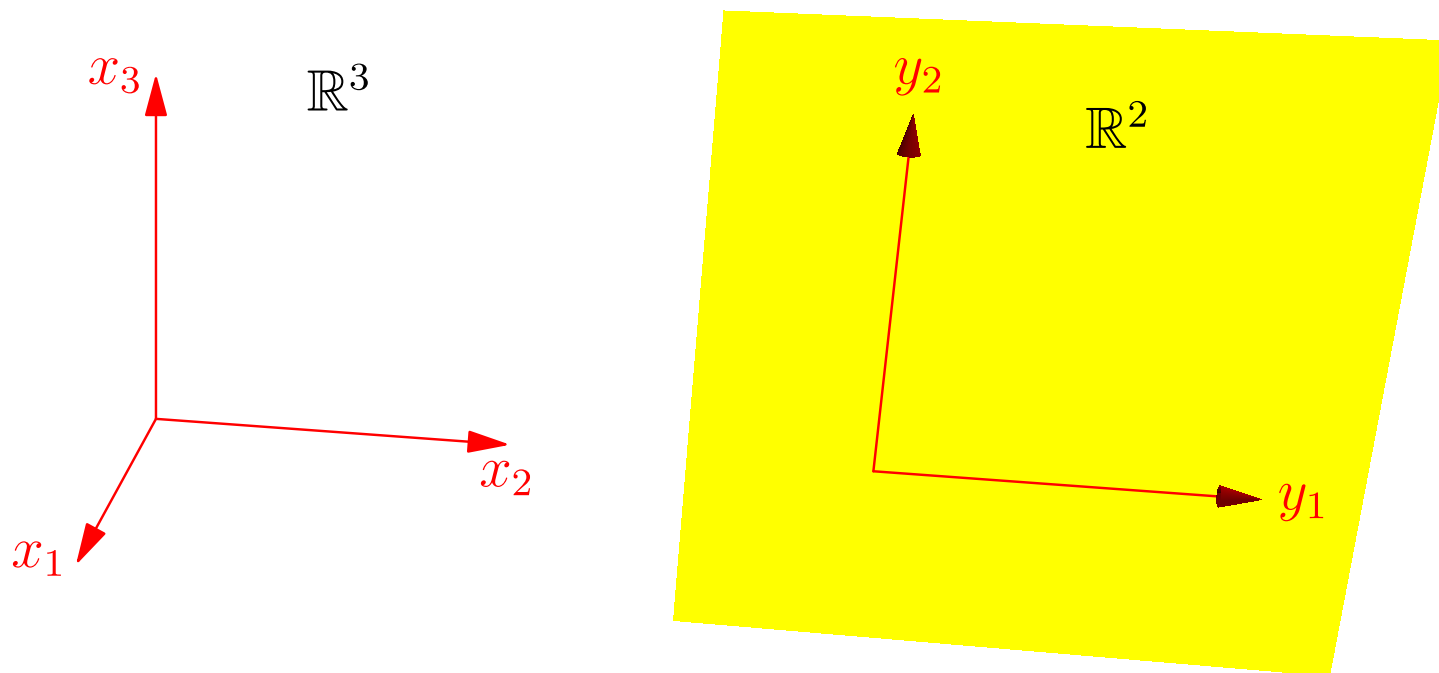
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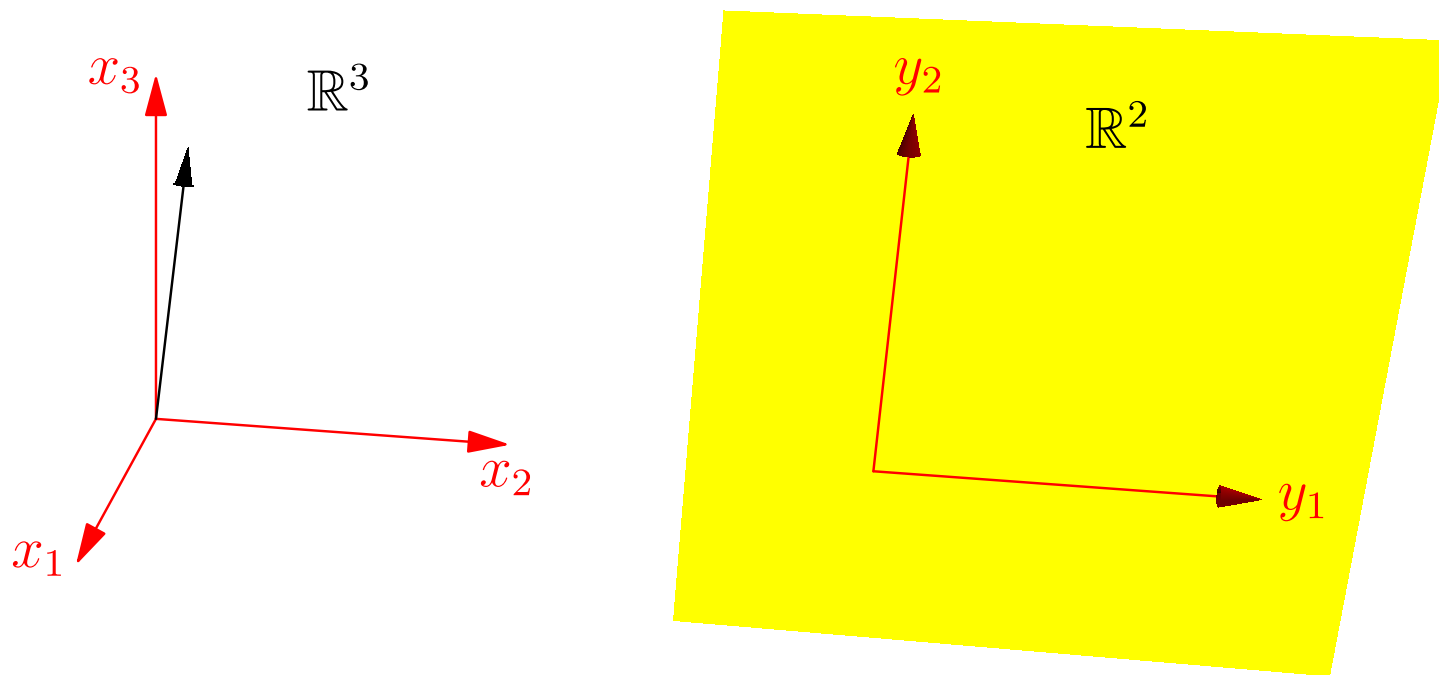
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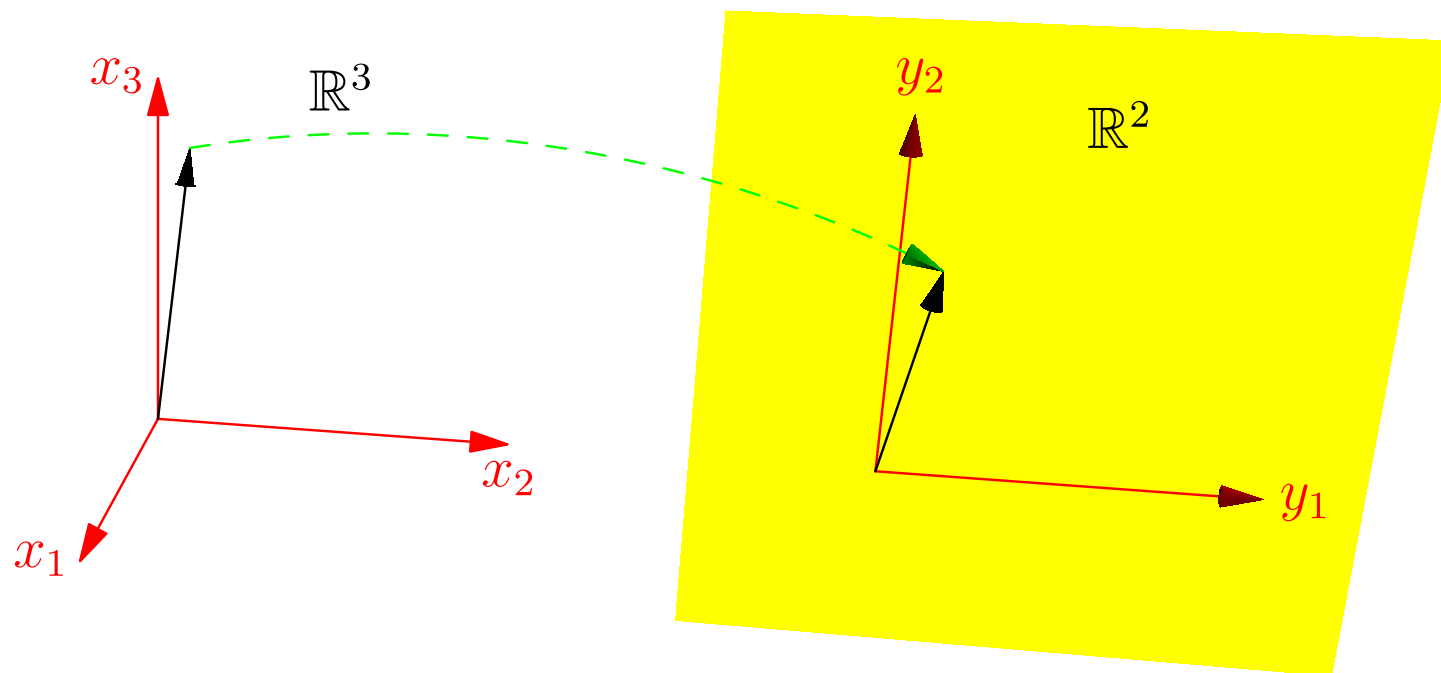
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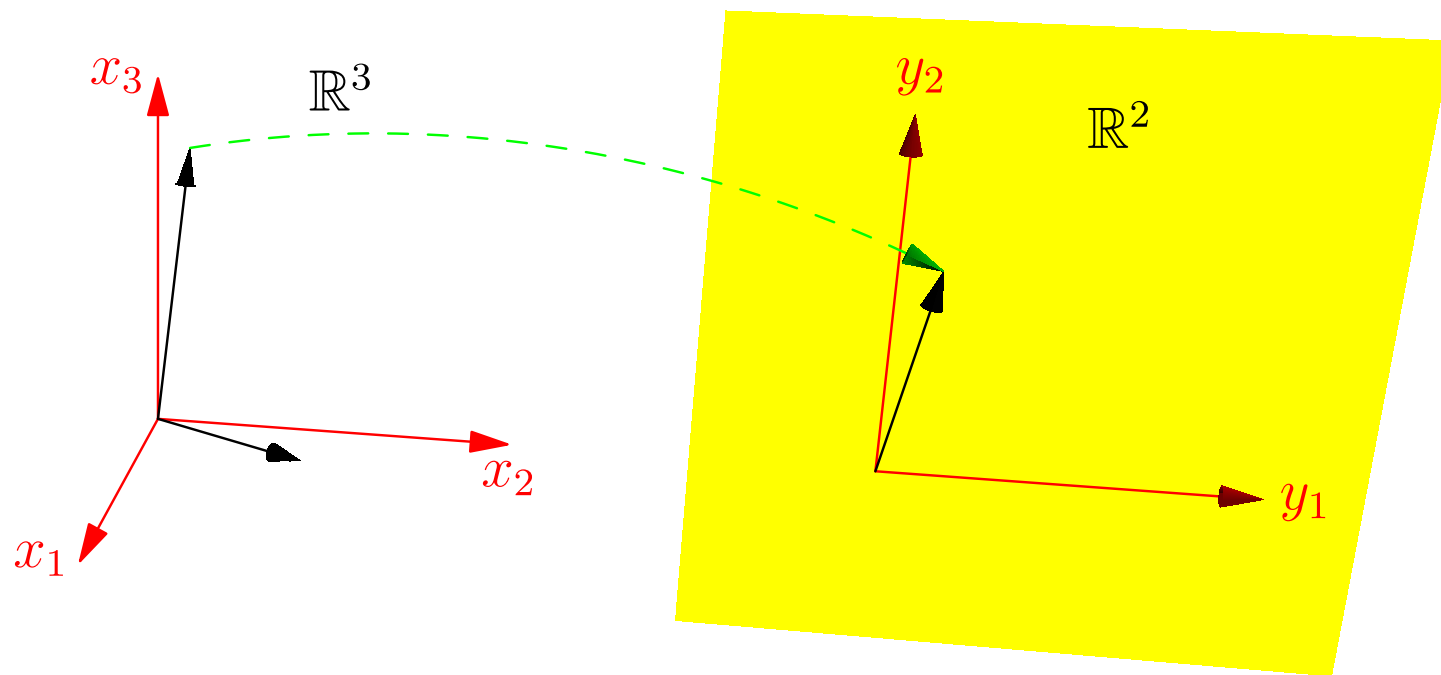
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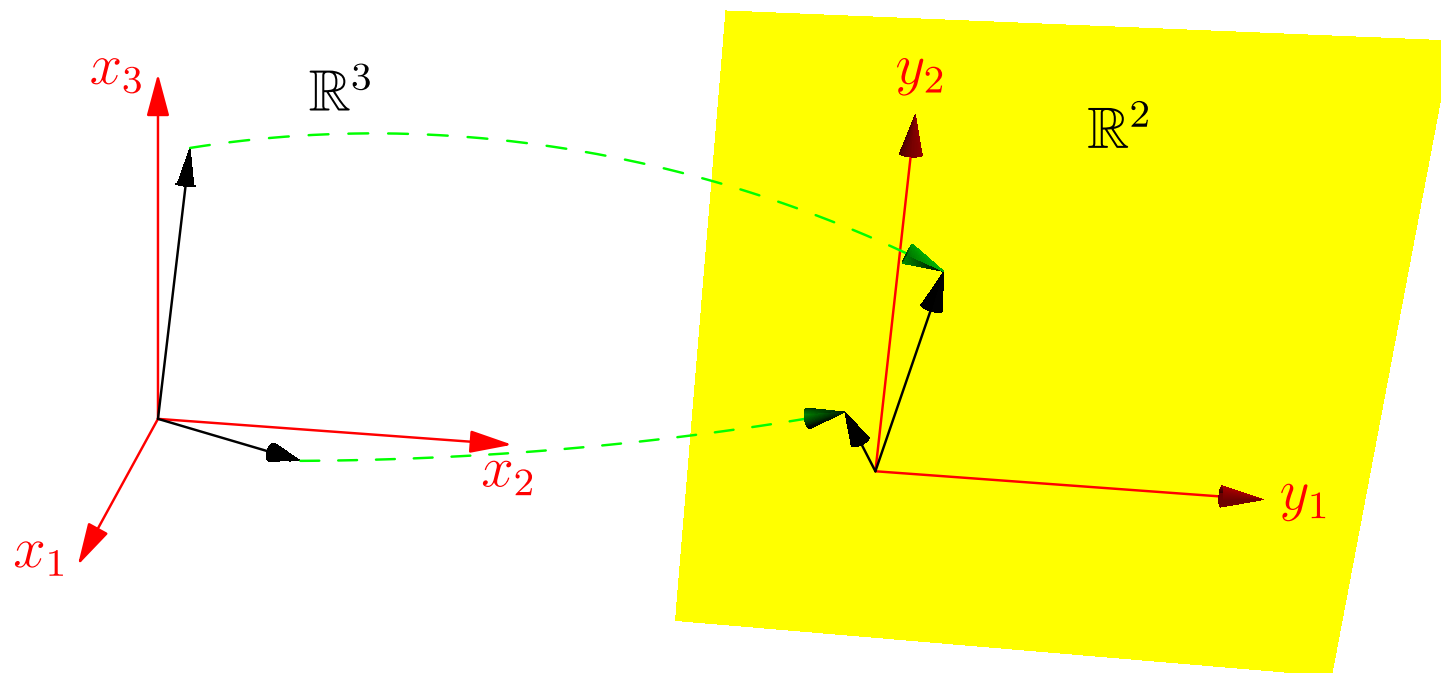
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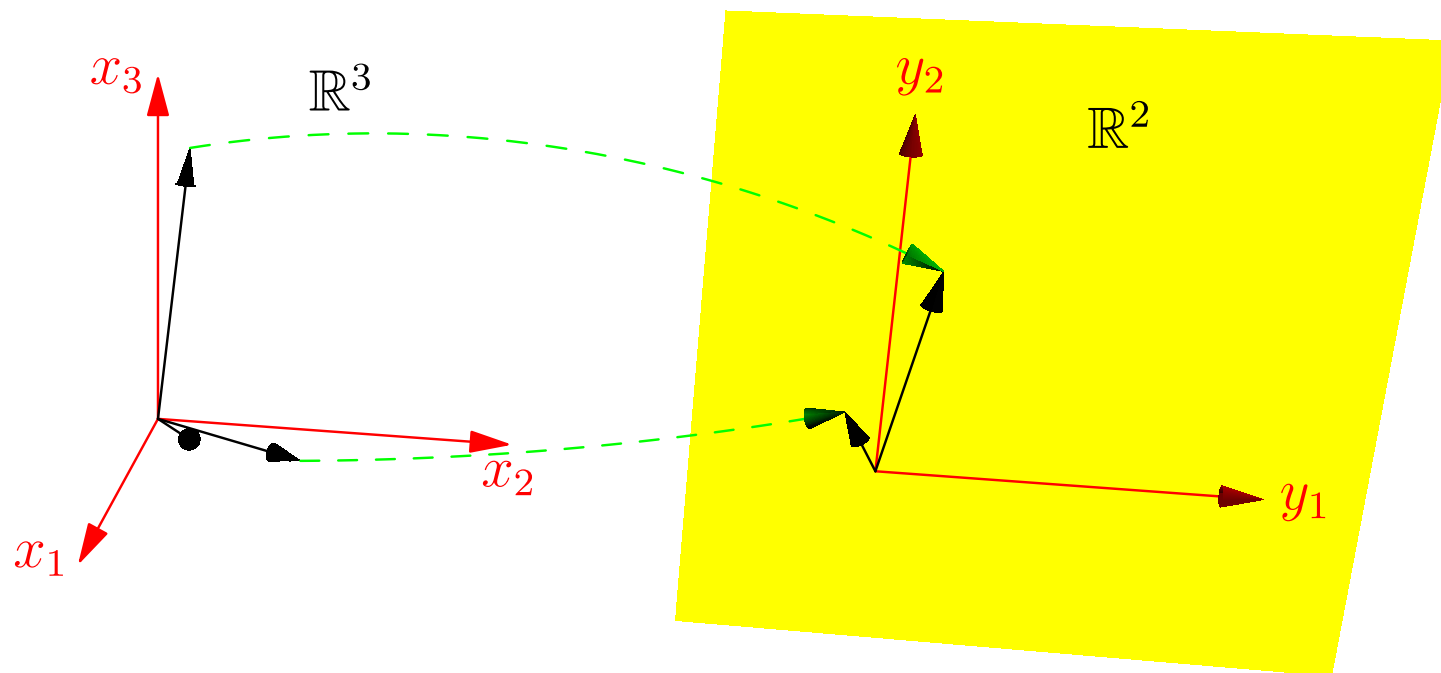
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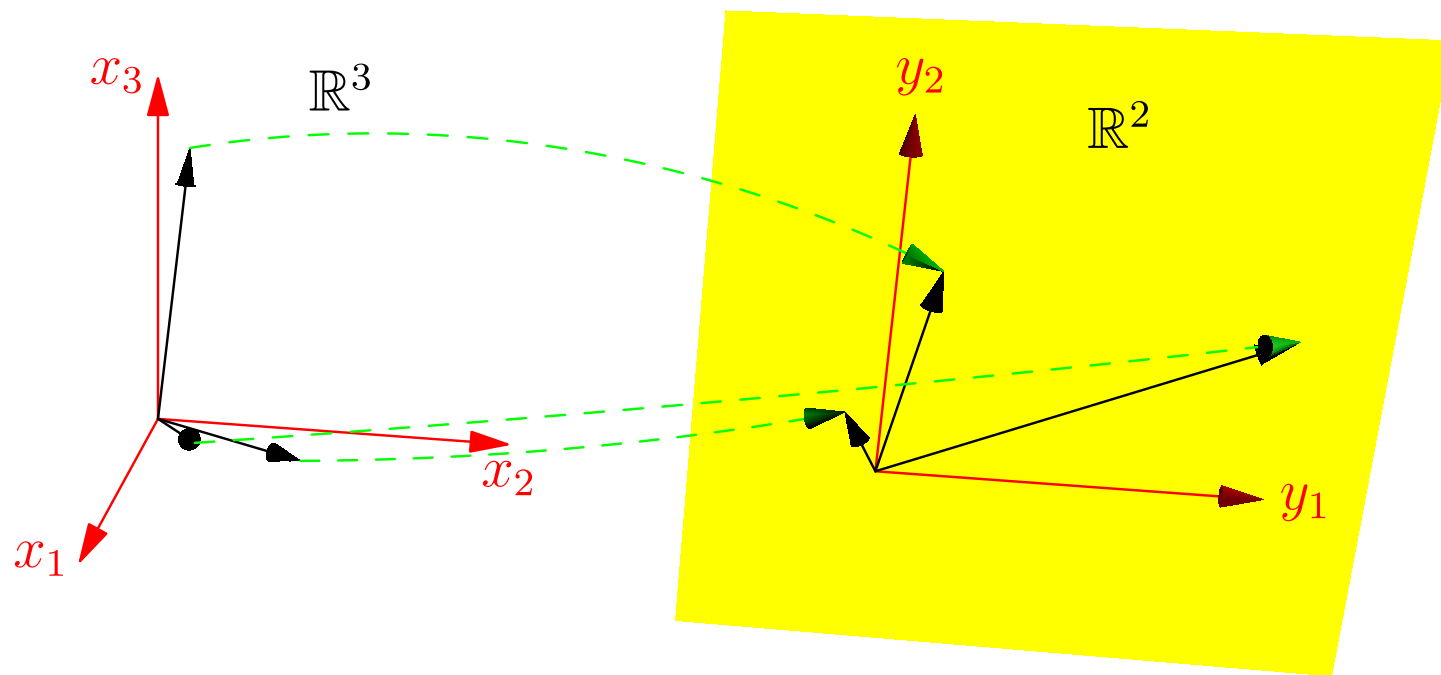
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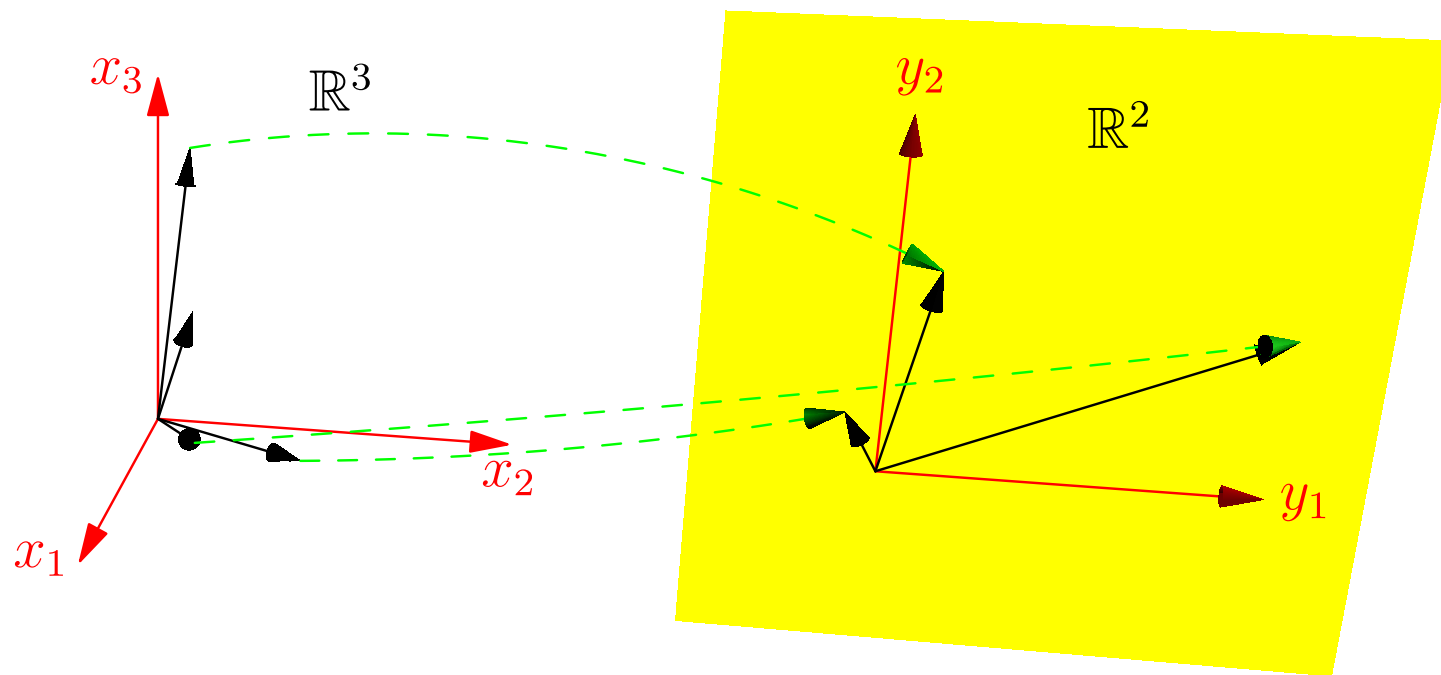
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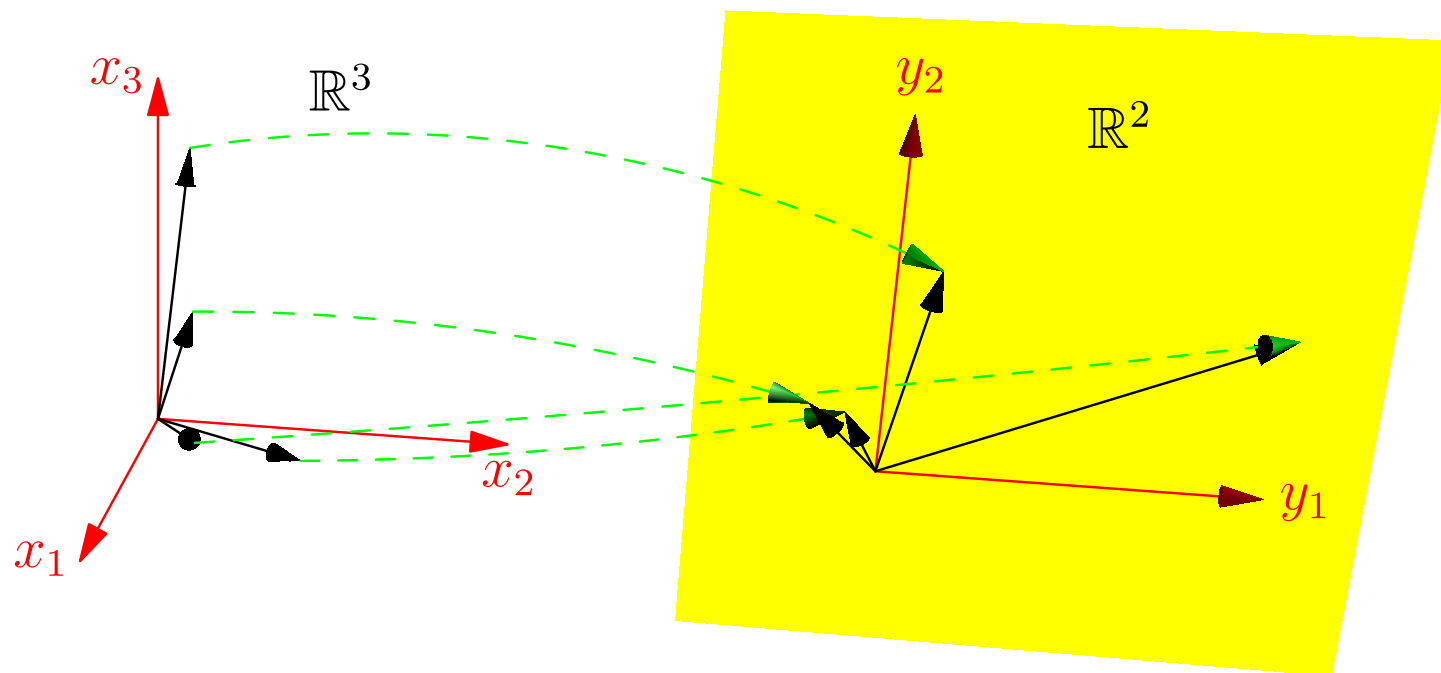
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